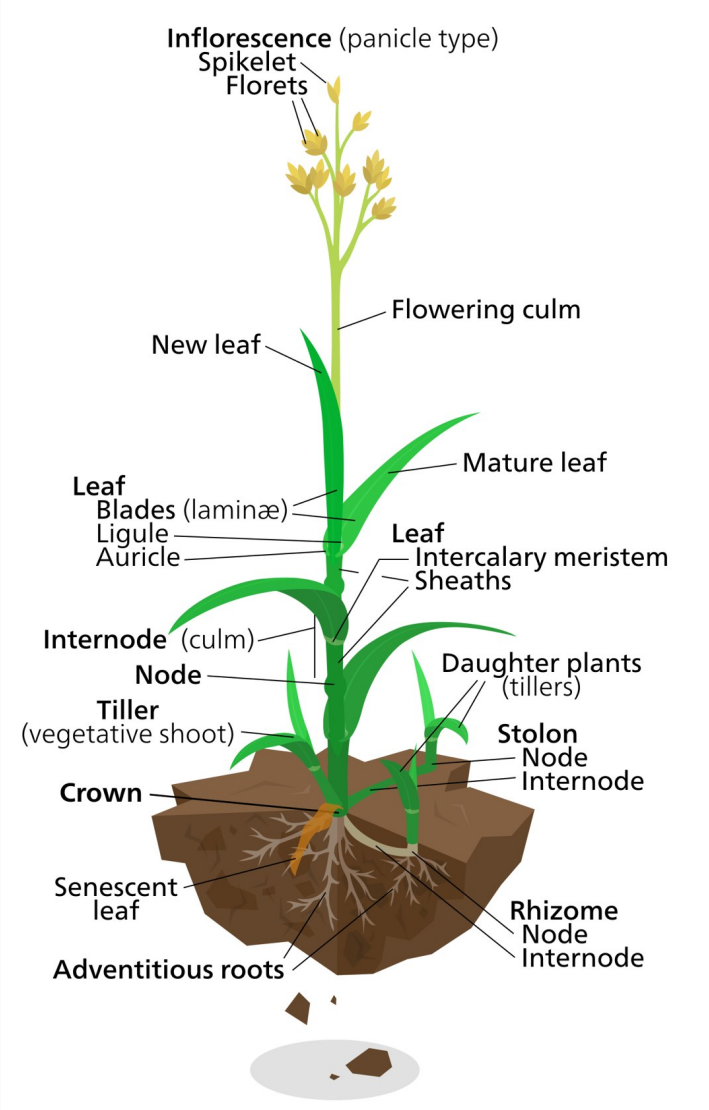
CITY OF KYLE / CONSERVATION CREW LEARNING

GRASS, UP CLOSE

A field workbook from beginner to advanced

See the whole grass. Trace a single leaf. Open the ligule. Follow the spikelets. Finally, learn the parts that separate look-alikes.



THE FIELD ROUTE

01 Stand back

02 Find the stem

03 Follow a leaf

04 Open the ligule

05 Read the head

06 Examine a spikelet

07 Check the base

Published art: Kelvinsong, CC BY 3.0. Not a particular native species.

Your first 10 minutes in the field

Start with a common grass at a trail edge or mowed bed. You do not need to know its name. Each stop moves closer, introduces a few terms, and asks you to point to the real structures.

1 LOOK	Stand 10-20 feet away. Describe shape and habitat.
2 FOLLOW	Choose one shoot and trace its leaves to the base.
3 OPEN	Bend one blade back to reveal the inside ligule.
4 ZOOM	Find one spikelet and compare the real diagram.
5 DOCUMENT	Photograph what you can see, mark what is still hidden.

Bring: hand lens (10x), small metric ruler, phone or camera, notebook, and a fresh grass sample where collection is permitted.

Important: Real anatomy examples may feature grasses that are not native to Hays County. We use their photographed structures as lessons, not as planting recommendations. A complete species ID requires multiple matching traits.

From beginner to advanced: read the large terms first, do the 'try it' task next, and only then read 'next level.'

BEFORE YOU START

Meet a grass. Do not name it yet.

A grass is a flowering plant. It has roots, stems, leaves, and tiny flowers. Today your job is to learn where those parts live before trying to identify a species.



Real grass in flower. The plant pictured is not identified to species; it teaches structure, not local native status.

WORDS TO LEARN

Blade

The long green part of a grass leaf. It collects sunlight.

Stem / culm

The supporting stalk that carries leaves and sometimes a flowerhead.

Seedhead / inflorescence

A cluster of tiny flowers or their developing fruits.

TRY IT OUTSIDE

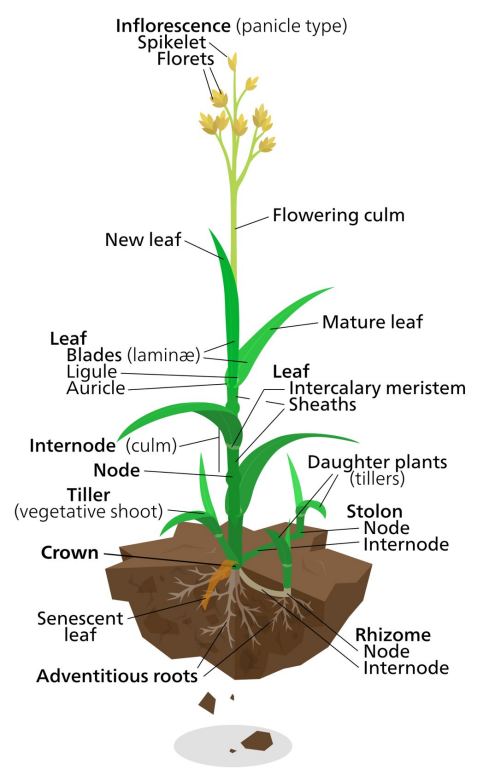
Stand 15-20 feet away. Point to the overall patch, individual plants, and one flowerhead. Say what you can actually see.

NEXT LEVEL

A plant with no seedhead can still be identified using leaves, sheaths, ligules and growth habit.

Read the whole plant's silhouette

Before you touch a grass, look at its shape. Shape gives useful clues, but it almost never proves the name by itself.



Published whole-plant diagram, Kelvinsong (CC BY 3.0). A generalized grass, not one species.

WORDS TO LEARN

Bunch / tuft

Many shoots stand close together, like a fountain from one base.

Mat / sod

Low stems spread outward and can form a carpet.

Habit

A botanist's word for the plant's overall growth form.

TRY IT OUTSIDE

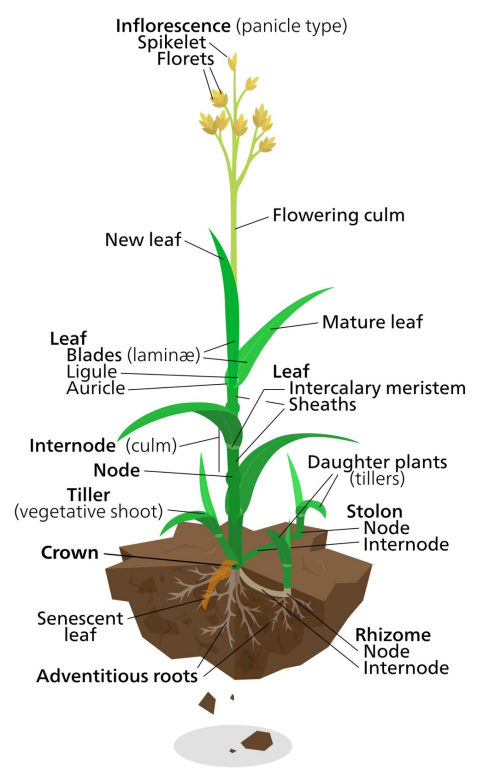
Ask: upright, arching, drooping, or creeping? Is it one clump, a patch, or a lawnlike mat? Sketch its outline.

NEXT LEVEL

Clumps may spread by short rhizomes; a dense patch does not automatically imply stolons. Check actual attachments.

Find one shoot in a clump

A single tuft can have many shoots. Follow just ONE upward. This is how you avoid mixing parts from different plants.



Kelvinsong's generalized grass includes a basal bud, daughter shoots and roots.

WORDS TO LEARN

Shoot

One developing leafy stem and its leaves.

Tiller

A new grass shoot that grows from a basal bud near the crown.

Crown

The tight region at soil level where shoots and roots connect.

TRY IT OUTSIDE

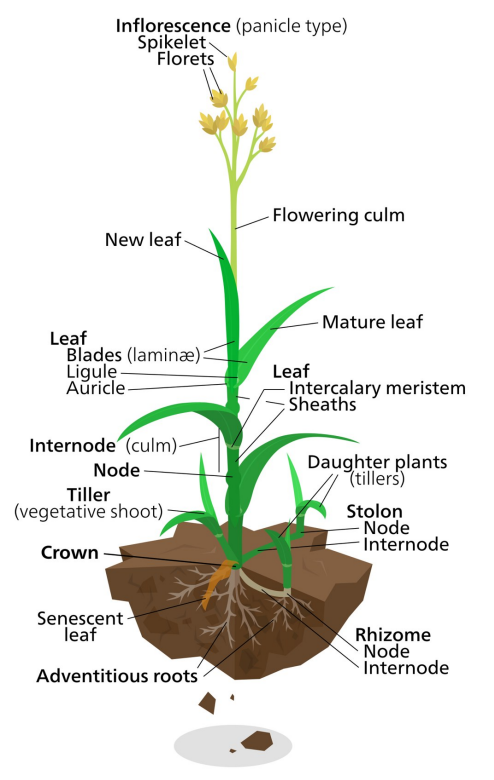
Pick a clump. Count visible shoots without pulling. Can you find younger leaves near the base?

NEXT LEVEL

A tiller can remain vegetative or eventually form a flowering culm. Cutting the flower stalk does not necessarily kill the crown.

Culm, node and internode

Think of a grass culm as a stalk built in segments. The junctions help you tell a grass stem from a loose leaf.



Look for horizontal joints on the drawn culm; the sheath may hide some nodes.

WORDS TO LEARN

Culm

The grass stem, usually round and often hollow between its joints.

Node

A joint on a culm. Grass leaves originate at nodes.

Internode

The stretch of stem BETWEEN two nodes.

TRY IT OUTSIDE

With your finger, trace a flowering stalk down to the first visible node. Count two internodes.

NEXT LEVEL

Many culms are hollow in their internodes but solid at nodes. Do not assume every grass culm is hollow.

A grass leaf has TWO big pieces

What we casually call one 'blade of grass' is actually a leaf made of an upper blade and a lower sheath.



Real Sorghum bicolor blade and sheath at their junction, photographed by Rasbak (CC BY-SA 3.0); this is an anatomy example, not a Hays native recommendation.

WORDS TO LEARN

Blade

The long, free part extending away from the stem; it may be flat, folded or rolled.

Sheath

The lower leaf portion that wraps around the culm.

Midrib / veins

The lengthwise support and smaller parallel lines running along many grass leaves.

TRY IT OUTSIDE

Find one green leaf. Trace its free blade DOWN until it starts wrapping around the stem. That wrap is the sheath.

NEXT LEVEL

Some sheaths are split; some are closed. Blade width, folding, hairs, and sheath opening all inform grass keys.

The collar is NOT the ligule

These are two nearby features that beginners often confuse. Look at both sides of the same leaf.



Real Cynodon dactylon leaf collar photograph with a scale; Stefan Lefnaer (CC BY-SA 4.0). The true inner ligule must be opened separately.

WORDS TO LEARN

Collar

The narrow band on the OUTSIDE of a grass leaf where blade meets sheath.

Ligule

A flap or row of hairs on the INSIDE of the blade-sheath junction.

Adaxial / abaxial

Inside/facing stem versus outside/facing away; useful technical terms when reading a key.

TRY IT OUTSIDE

Hold the blade away from the stem. Turn it over to inspect the inside seam. Don't call outer sheath hairs the ligule.

NEXT LEVEL

Ligules can be membranous, hairy, mixed, very short or absent. Different leaf ages can appear different.

A membranous ligule looks like a flap

A membrane is a thin sheet of tissue. Imagine a tiny translucent fingernail standing up inside the leaf base.



Rasbak's actual Sorghum bicolor inner ligule photograph (CC BY-SA 3.0).

WORDS TO LEARN

Membranous ligule

A thin, often whitish or translucent flap at the inside junction.

Margin

The top edge of that flap. It may be smooth, jagged or hairy.

Height

How far the ligule projects upward; sometimes a fraction of a millimeter.

TRY IT OUTSIDE

Gently pull the blade back. Look at its inner base against light. If you have a 1-mm ruler, photograph it beside the flap.

NEXT LEVEL

Measure from the sheath mouth to the tip of the ligule - not from the nearest visible sheath hair.

A hairy ligule is a FRINGE

Not every grass makes a membrane. Some make a tiny row of hairs. The PHOTO shows an exterior collar: you still need to turn the leaf back to locate the true inner ligule.



Cynodon dactylon (Bermudagrass) external collar and hairs, Stefan Lefnaer (CC BY-SA 4.0); the very short inner ligule is not independently exposed.

WORDS TO LEARN

Ciliate / cilia

Fringed with tiny hairs.

Fringe of hairs

Short hairs along the INSIDE edge where blade joins sheath.

Pubescent

A word for a plant surface covered with hairs; not necessarily a ligule.

TRY IT OUTSIDE

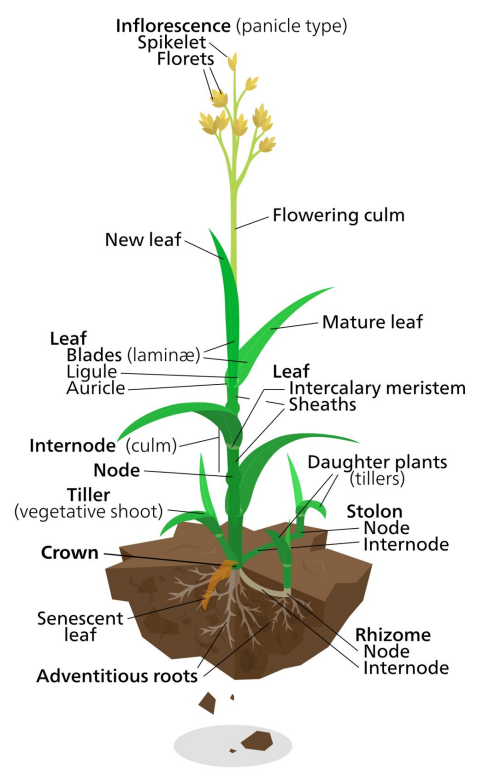
Compare an inner ligule with hairy blade surfaces or sheath margins. Are the hairs exactly at the interior joint?

NEXT LEVEL

In sand lovegrass, conspicuous external collar hairs must not be mistaken for the very short true internal ligule.

Auricles and how a new leaf opens

Sometimes the leaf base has two tiny 'ears' that grab the stem. Young leaf blades can also arrive folded or rolled.



General grass form, Kelvinsong, CC BY 3.0. Auricle detail varies by species.

WORDS TO LEARN

Auricles

Small ear-shaped projections at the blade base; many grasses lack them.

Vernation

How a new blade sits in the unopened shoot: folded or rolled.

Sheath mouth

Where the sheath ends and the free blade begins.

TRY IT OUTSIDE

Check BOTH edges of the leaf base for auricles. Look at a young shoot before unrolling it.

NEXT LEVEL

No auricles is a useful observation, not a diagnosis. Vernation requires an intact young shoot.

A seedhead is a WHOLE cluster

One 'seedhead' may hold dozens or hundreds of tiny flowering units. Look at its branching pattern before one small spikelet.



Unidentified photographed grass with densely clustered flowerheads, RoRo (CC0); a field structure demonstration only.

WORDS TO LEARN

Inflorescence

The entire flowering structure of a plant.

Panicle

A branching inflorescence: side branches arise from a main axis.

Rachis

The main axis or backbone of a flowering structure.

TRY IT OUTSIDE

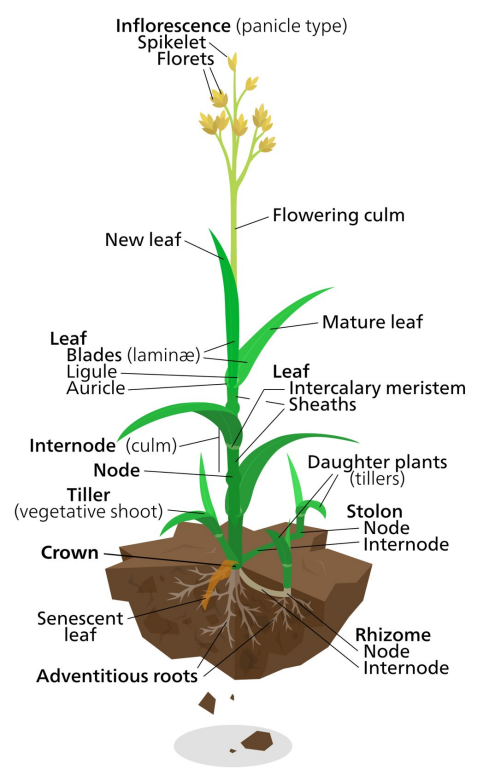
Trace the head from the stalk to the main axis. Find one branch and see whether smaller units attach to it.

NEXT LEVEL

A contracted panicle can look spikelike; move branches apart gently and check attachment.

Spike, raceme, and panicle

These words explain HOW the tiny units are arranged. Think: direct attachment, short stalk attachment, or branches upon branches.



The generalized whole-plant diagram shows only ONE possible inflorescence pattern.

WORDS TO LEARN

Spike

Spikelets attach to a main axis without their own stalks.

Raceme

Spikelets attach to an axis on little stalks (pedicels).

Panicle

Branching axis with smaller side branches, often airy and open.

TRY IT OUTSIDE

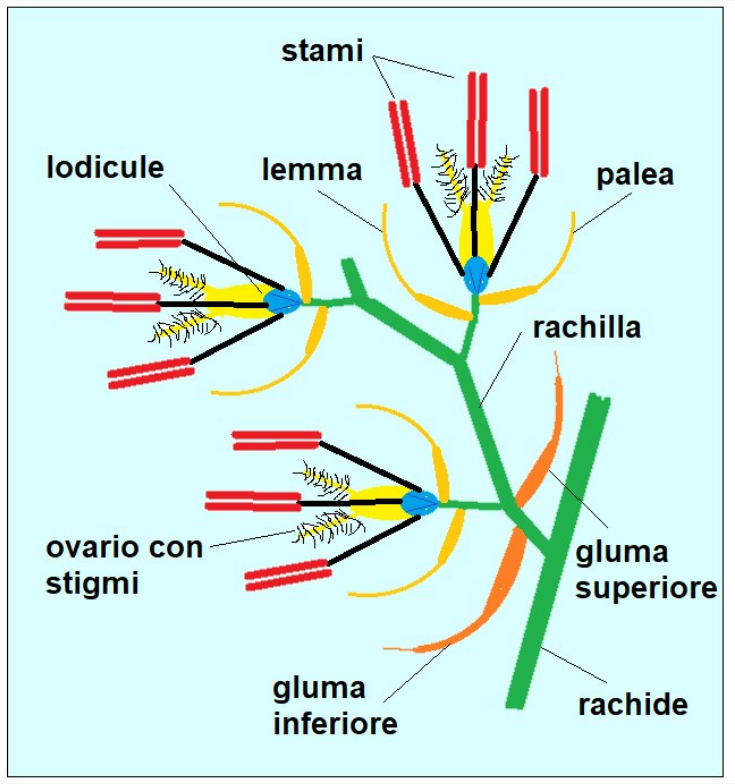
Observe an intact head from the side. Draw a single axis plus its branches. Don't call a head a spike just because it looks narrow.

NEXT LEVEL

Certain bluestems bear paired sessile and pedicelled spikelets on specialized branches, sometimes called rames.

What is a spikelet?

A spikelet is the small grass reproductive unit you can often see with the naked eye. It is not always one flower.



Existing published spikelet illustration by Enrico Blasutto (CC BY-SA 4.0), with original Italian labeling; translate with the English glossary.

WORDS TO LEARN

Spikelet

The little package that holds one or more grass florets.

Glumes

Usually two lower bracts at the base of a spikelet, though some grasses differ.

Rachilla

The small internal axis that bears the florets inside a spikelet.

TRY IT OUTSIDE

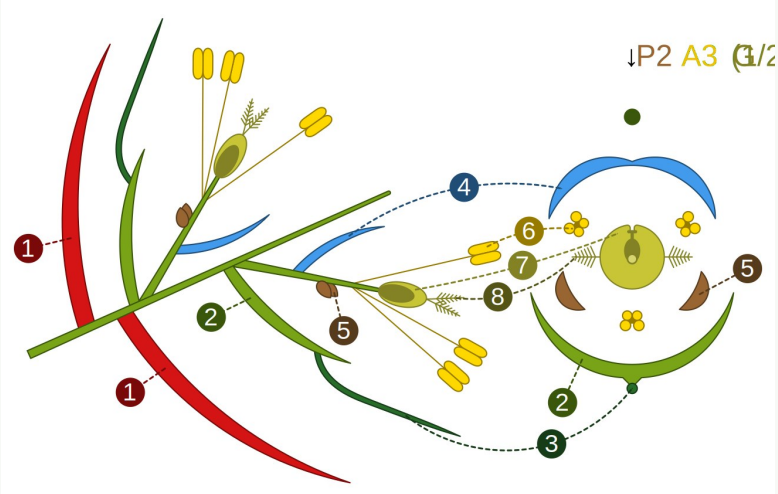
Pick one intact spikelet and sketch its overall shape. Can you find the two basal bracts? Do NOT pull it apart yet.

NEXT LEVEL

Species identification may depend on whether glumes are equal, hairy, keeled, absent, or longer than the florets.

One floret is one tiny flower

Grass flowers usually do not have colorful petals. Their protective bracts make them look like miniature overlapping scales.



Numbered generalized grass-flower teaching schematic. We attempted to retrieve the published CC0 diagram first; its source link appears in the credits.

WORDS TO LEARN

Floret

A grass flower plus its two surrounding bracts.

Lemma

The outer protective bract of a floret.

Palea

The usually smaller inner protective bract of a floret.

TRY IT OUTSIDE

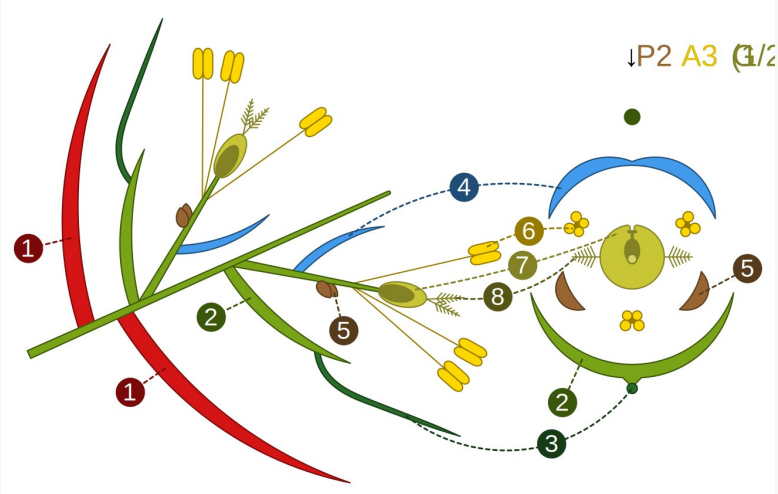
First identify the WHOLE spikelet. Then trace inward to a floret. Remember: glumes are usually outside and below the florets.

NEXT LEVEL

The lemma and palea are not petals. Their texture, nerves, teeth and awn points are useful at species level.

Anthers carry pollen; stigmas catch it

A grass is a flowering plant even if the flowers look like specks. What emerges from florets can be extremely diagnostic.



Numbered botanical teaching schematic with eight labeled structures; original public-domain reference is linked in source notes.

WORDS TO LEARN

Anther

The pollen-producing part, often a dangling yellow, purple or brown little sac.

Stigma

The pollen-catching part, often feathery and branched.

Ovary

The structure inside the flower that can mature into the fruit.

TRY IT OUTSIDE

Use the diagram to find #6 anthers, #7 ovary and #8 feathery stigmas. Look for the same organs on a fresh flowering head.

NEXT LEVEL

Grass flowers commonly release pollen to the wind. Flowering stage changes how easily the structures can be photographed.

Translate the flower diagram

Here is the complete key for the eight numbers on the public-domain diagram. Use this after you understand what a spikelet and floret are.

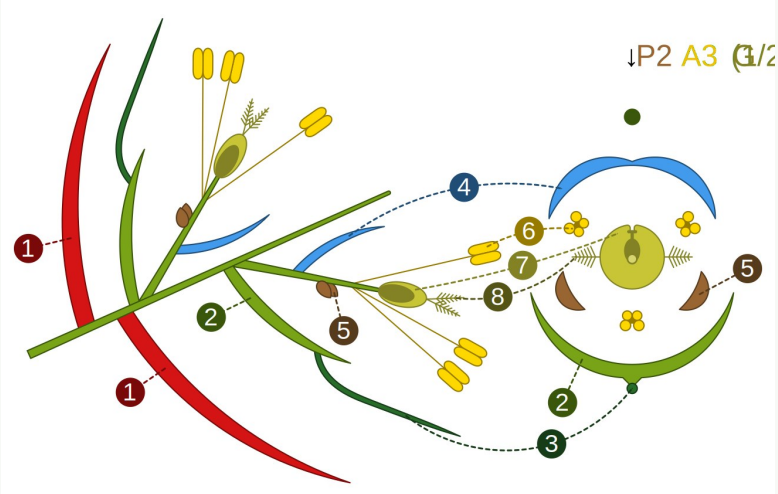


Diagram used to teach the same scientific structures; external reference and fallback authorship are identified in source notes.

WORDS TO LEARN

1 Glumes | 2 Lemma

The outer spikelet bracts and the outer floret bract.

3 Awn | 4 Palea

A narrow projection (if present), and the inner floret bract.

5 Lodicules | 6 Stamens

Tiny scales associated with opening the flower, and pollen organs.

7 Ovary | 8 Stigmas

Seed-producing floral structure and the feathery pollen-catching ends.

TRY IT OUTSIDE

Say these words while pointing to actual numbered structures. Which ones can you see without magnification?

NEXT LEVEL

Floret orientation and number vary. This diagram is an educational generalized grass example, not a species key.

Awn, bristle, or an ordinary hair?

These three structures can all look like thin threads, but they are not the same thing. Their attachment is the clue.



Real Sorghum bicolor spikelet-group macro, Matt Lavin (CC BY-SA 2.0), archived in the Hays Botany repo.

WORDS TO LEARN

Awn

A stiff or slender projection growing from a glume or lemma.

Bristle

A stiff hairlike structure that may develop on an axis below or near spikelets.

Hair / trichome

A small outgrowth from an epidermal surface (stem, sheath, blade or bract).

TRY IT OUTSIDE

If you spot a thread, follow it to its BASE. Does it arise directly from a floral bract, or beside an entire spikelet?

NEXT LEVEL

In Setaria bristles arise from modified branches below the spikelets; in Aristida the awns belong to the lemma.

Roots are not the same as rhizomes

Roots absorb water and anchor a grass. Some grasses also have true horizontal STEMS that run beneath the soil.



Real two-week-old Sorghum bicolor seedlings with adventitious roots, Miquel Pujol Palol (CC BY-SA 3.0).

WORDS TO LEARN

Fibrous roots

Many narrow roots, rather than one large woody taproot.

Crown

The place where the shoots and attached roots meet.

Rhizome

A STEM growing under soil. Look for nodes, scale leaves, buds and shoots, not just roots.

TRY IT OUTSIDE

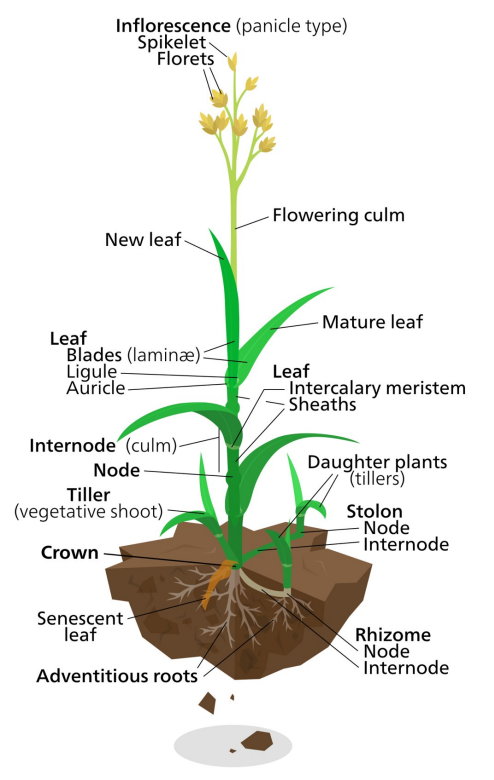
At an approved site, gently clean a collected specimen and photograph shoot bases connected to roots. Don't collect in protected areas.

NEXT LEVEL

A young seedling root photo does not prove the architecture of a mature perennial crown; follow the entire plant connection.

Stolons run on top; rhizomes run below

When a grass forms a patch, learn how new shoots connect. This often matters more than the seedhead color.



Kelvinsong's published generalized diagram includes above-ground stolon and below-ground rhizome.

WORDS TO LEARN

Stolon

Horizontal stem that runs at or above ground and can root at nodes.

Rhizome

Horizontal underground stem with joints and buds.

Tiller

A new shoot from a basal bud; not automatically a runner.

TRY IT OUTSIDE

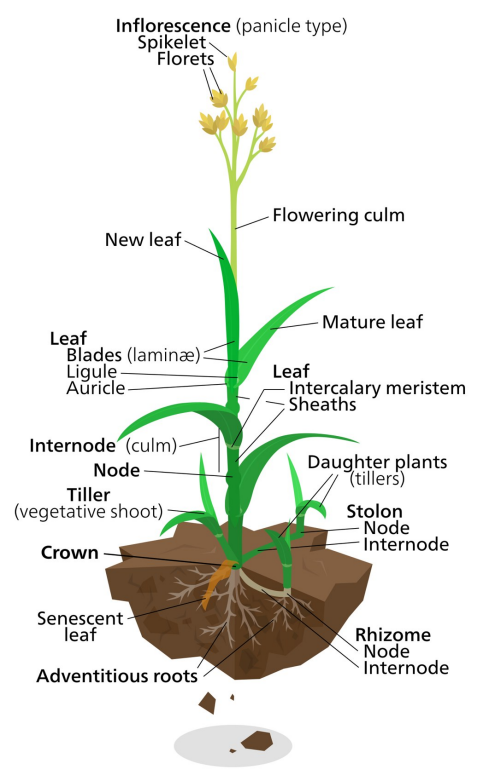
Point to one grass shoot. Try to trace any horizontal connecting stem. Photograph actual nodes before calling it a stolon or rhizome.

NEXT LEVEL

Johnsongrass has characteristic spreading rhizomes; typical crop sorghum lacks that perennial network. Do not spread fragments.

A species name needs several clues

One shape or color is not enough. A good identification uses multiple independent structures on the SAME specimen.



Generalized published plant illustration (CC BY 3.0).

WORDS TO LEARN

Vegetative characters

Features of leaves, sheaths, ligules, stems and roots.

Reproductive characters

Features of flowerhead, spikelets and florets.

Look-alike

A different species with similar overall appearance.

TRY IT OUTSIDE

Write three things you observed and one thing you still need to photograph. Keep the name tentative until evidence agrees.

NEXT LEVEL

Keys may use glume nerves, lemma texture, awn insertion, ligule size, rhizome nodes and spikelet disarticulation.

Grass, sedge or rush?

Some grasslike plants are not actually grasses. The common shortcut about stem shape is helpful but has exceptions.



General photo for observation; grasslike silhouettes alone do not settle plant family.

WORDS TO LEARN

Grass (Poaceae)

Usually has a sheath, a ligule region, jointed culms, and tiny spikelets.

Sedge (Cyperaceae)

Often has 3-angled, solid culms; leaves may be in three ranks.

Rush (Juncaceae)

Often has round solid stems and small flower parts, not grass spikelets.

TRY IT OUTSIDE

Before keying a 'grass,' inspect the leaf sheath and flowers. Do not rely ONLY on 'sedges have edges' or stem feel.

NEXT LEVEL

Some sedges have round stems, and some grasses have solid culms. Reproductive organs are the decisive test.

Photograph a grass in five steps

A clear photo sequence makes someone else able to check your ID even after you leave the park.



Licensed Cynodon collar-hair image, Stefan Lefnaer (CC BY-SA 4.0), shows useful framing; it does not prove inner ligule length.

WORDS TO LEARN

1. Habitat & habit

Photograph the patch, its soil and its growth form.

2. Whole individual

Shoot from crown to top of flowering head.

3. Collar & ligule

Photograph outside first, then peel back and show the INSIDE with a scale.

4. Inflorescence

Photograph attached branching pattern.

5. Spikelet & base

Photograph close-up reproductive parts and roots/runners if available.

TRY IT OUTSIDE

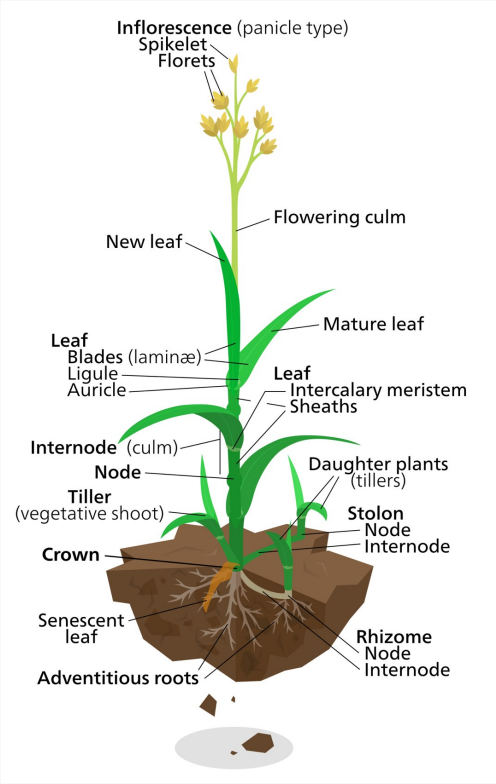
Try this on one common park grass. Keep at least one image with a ruler or coin and write the location.

NEXT LEVEL

Take original unedited images first. When adding arrows, label only anatomically visible evidence. Keep species verification separate from photo rights.

Can your crew find six structures?

Use one fresh grass plant near the trail. Work in pairs; the goal is correct vocabulary before a species guess.



Generalized grass reference by Kelvinsong (CC BY 3.0).

WORDS TO LEARN

A. From 10 feet

Bunch or mat? Upright or arching? Presence of seedhead?

B. From 2 feet

Locate stem, node, blade and sheath.

C. With your fingers

Find outside collar; turn leaf back to find inner ligule.

D. With 10X lens

Choose one spikelet and identify at least one bract.

TRY IT OUTSIDE

Partner A points to a part; Partner B names it, explains it in ordinary words, and says whether it is visible.

NEXT LEVEL

If a part is hidden, record 'not seen' instead of guessing. Unknown is a good field observation.

One real grass: field observation sheet

Do this WITHOUT looking up a name. Record actual evidence, then use a flora or key. A blank answer is better than an invented part.

Park / date / observer

Habitat: dry slope, open lawn, riparian etc.

Habit: bunch, mat, single stalk, unknown

Culm and node: where did you find it?

Blade: flat, folded, rolled, hairs?

Sheath and OUTSIDE collar: description

INSIDE ligule: membrane, hairs, absent, unseen

Auricles: present, absent, unseen

Inflorescence: spike, raceme, panicle, unclear

Spikelet: size / number of florets / glumes

Base: roots, crown, stolon, rhizome, unseen

Tentative ID & two look-alikes to eliminate

SELF CHECK

Quiz: from observation to diagnosis

01 What is the difference between a blade and its sheath?

02 Does a collar sit mainly on the outside or inside of a leaf?

03 Which structure is the little flap or hair fringe on the inside?

04 A culm has three visible nodes. What are the spaces between them called?

05 What is the difference between a whole flowerhead and a spikelet?

06 Which bract belongs to a floret: lemma or glume?

07 What does a feathery grass stigma do?

08 Where does a stolon usually grow compared with a rhizome?

09 Why is a fringe of hairs on the OUTSIDE sheath not proof of a hairy ligule?

10 Your photo shows a panicle but no ligule. What should you report?

Ten answers worth remembering

1. Blade is free, usually long and narrow; sheath is the lower wrapping leaf portion.

2. Collar is on the outer surface where blade and sheath meet.

3. The ligule: on the inside, sometimes membrane, hairs or apparently absent.

4. Internodes.

5. Inflorescence is the whole flowering structure; spikelet is one small unit within it.

6. Lemma belongs to a floret; glumes are typically at the spikelet base.

7. To catch pollen; it may be feathery to collect windborne grains.

8. Stolon usually runs at/above soil, rhizome below soil; both are stems.

9. Location matters: only hairs at the inner blade-sheath seam constitute a hairy ligule.

10. Record 'ligule not seen' and take another photo; do not claim a feature without evidence.

GOOD FIELD HABIT: When unsure, record the trait as unknown. A reliable ID can be checked by someone else.

External illustrations and photographs

This edition replaces most generic drawings with published scientific illustrations and real species-source photographs. A simplified flower schematic is retained as a clearly marked fallback. Original image creators retain credit. Crop/grassy images illustrate structures, not native plant status.

WHOLE | Kelvinsong / CC BY 3.0
https://commons.wikimedia.org/wiki/File:Grassy_grass_plant.svg

SPIKELETLET | Enrico Blasutto / CC BY-SA 4.0
https://commons.wikimedia.org/wiki/File:Poaceae_-_Spighetta.png

FLORAL | Nefronus, Alfred, Enzoklop / CC0
https://commons.wikimedia.org/wiki/File:Grassbluete_numbers_clones_unlinked.svg

FIELD | RoRo / CC0
https://commons.wikimedia.org/wiki/File:Poaceae_-_unidentified_plant_and_spikelets.jpg

HAIR | Stefan Lefnaer / CC BY-SA 4.0 (true Cynodon collar region, 2-mm photo scale)
https://raw.githubusercontent.com/Crking92/Hays-County-Botany/main/content/taxa/Poaceae/C...

MEM | Rasbak / CC BY-SA 3.0 (actual Sorghum bicolor inner ligule)
https://raw.githubusercontent.com/Crking92/Hays-County-Botany/main/content/taxa/Poaceae/S...

SORGHUM ROOTS | Miquel Pujol Palol / CC BY-SA 3.0
https://raw.githubusercontent.com/Crking92/Hays-County-Botany/main/content/taxa/Poaceae/S...

SORGHUM SPIKELET | Matt Lavin / CC BY-SA 2.0
https://raw.githubusercontent.com/Crking92/Hays-County-Botany/main/content/taxa/Poaceae/S...

License details: CC BY requires attribution; CC BY-SA also requires ShareAlike for adapted material. CC0 is public-domain-dedicated. Photographs are reused as examples; labels and teaching explanations in this PDF are new.

Where to learn even more

Montana State Extension - Grass Identification Basics

https://extension-store.montana.edu/montguides/grass-identification-basics

Earth@Home - Grass Morphology & Anatomy

https://evolution.earthathome.org/grasses/morphology/

Oregon State - Grass Structures

https://forages.oregonstate.edu/regrowth/how-does-grass-grow/grass-structures

USDA - Typical Grass Structures

https://www.nrcs.usda.gov/plantmaterials/etpmcpu8052.pdf

Hays Botany Learner (regional grass study)

https://crking92.github.io/Hays-County-Botany/

Final question: can you identify a tiny grass structure without a photograph showing that structure? No. The most important advanced skill is knowing what you have actually observed.